apples and 4 units of oranges, so that he can get (88+54) 142 units of total utility.

Here $\frac{MU_A}{P_A} = \frac{MU_o}{P_O}$ ie, $\frac{10}{2} = \frac{5}{1} = 5$

Diagrammatic Illustration

In diagram 2.2, X axis represents the amount of money spent and Y axis represents the marginal utility of Apple and Orange respectively. If the consumer spends ₹10 on Apple and ₹4 on Orange, he will buy five apples and four oranges. The marginal utilities of both Apple and Orange are equal i.e.,AA_1=BB_1 (5=5). Hence, he gets maximum utility.

$A_2 A_3$ and $B_2 B_3$ lines have not been used for explanation.

Criticisms

1. In practice, utility cannot be measured, only be felt.
2. This Law cannot be applied to durable goods.

2.7 Consumer's Surplus

The concept of consumer surplus was originally introduced by classical economists and later modified by Jevons and Jule Dupuit, the French Engineer Economist in 1844. But a most refined form of the concept of consumer surplus was given by Alfred Marshall. This concept is based on the Law of Diminishing Marginal Utility.

Definition

Alfred Marshall defines consumer's surplus as, "the excess of the price which a person would be willing to pay rather than go without the thing, over that which he actually does pay is the economic measure of this surplus satisfaction. This may be called consumer's surplus".

Assumptions

1. Marshall assumed that utility can be measured.
2. The marginal utilities of money of the consumer remain constant.
3. There are no substitutes for the commodity in question.
4. The taste, income and character of the consumer do not change.
5. Utility of one commodity does not depend upon the other commodities.

Explanation

The concept of consumer's surplus can be explained with help of an example. Suppose a consumer wants to buy an apple. He is willing to pay ₹4, rather than go without it and the actual price of the apple is ₹2. Hence the consumer's surplus is ₹2(₹4 – ₹2). Thus, consumer's surplus is the difference between the price that a consumer is willing to pay (potential price) and what he actually pays. Therefore,

Consumer's surplus	=	What a person is willing to pay – What he actually pays.
	OR	
Consumer's surplus	=	Potential price– Actual price.

Mathematically,

Consumer's surplus = TU – (P x Q)

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